

# 1. Teacher CBT CA Configuration (The "Precision" Module)

*Goal: Allow teachers to create assessments that match the specific cognitive needs of the class.*

- [ ] **Difficulty Mapping:** Can the teacher assign tags (Easy, Medium, Hard) to questions and filter the test to include a specific ratio (e.g., 60% Easy, 40% Hard)?
- [ ] **Timed Logic:** Does the timer sync across devices? If a student refreshes the browser, does the countdown pick up from the remaining time or reset? (It must pick up).
- [ ] **Question Scoping:** Can the teacher toggle between "Topic-Specific" (e.g., *only* Quadratic Equations) and "General" (full-term review)?
- [ ] **Randomization Toggle:** A "Shuffle" function for both question order and option order to prevent student collaboration.
- [ ] **Instant Review:** Option for the teacher to set whether students see their score immediately or wait until the Admin releases it.

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# 2. Admin Grade Scale Customization (The "Authority" Module)

*Goal: Give the school principal total control over their academic standards.*

- [ ] **Dynamic Range Mapping:** A UI where the admin can set the boundaries.
  - *Example: \$75 - 100 = A\$, \$60 - 74 = B\$.*
- [ ] **Remark Personalization:** Can the admin edit the automated remarks (e.g., "Excellent," "Distinction," "Needs Improvement") to match the school's "voice"?
- [ ] **Weighting Configuration:** Can the admin decide the ratio of CA to Exam? (e.g., 40% CA, 60% Exam).
- [ ] **Global Apply:** A "Save & Apply" button that retroactively updates all broadsheets in the system when the scale is changed.

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# 3. Broadsheet Data Validation (The "Clean Data" Guardrail)

*Goal: Prevent "Manual Chaos" by ensuring only valid, numerical data enters the system.*

- [ ] **The "Float-Only" Filter:** Implement a regex or input mask that prevents letters or special characters (e.g., "A", "-", "5.5.") from being saved.
- [ ] **Visual Flagging:** If an invalid entry is attempted, the cell must glow red (`.r-1oifz5y`) and display a tooltip: *"Numerical values only."*
- [ ] **Boundary Check:** Any number outside the \$0 - 100\$ range must be automatically rejected with a prompt: *"Score cannot exceed the maximum mark."*

- [ ] **Auto-Calc Integrity:** Does the "Total" column update in real-time (8-second loop) as the teacher types, or does it require a page refresh? (It must be real-time).

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## 4. Report Card Download Engine (The "Output" Module)

*Goal: Generate professional, branded documents that parents can't wait to share.*

- [ ] **PDF Formatting:** Ensure the report card fits a standard A4 page perfectly without cutting off rows or the school logo.
  - [ ] **Batch Download:** Can the admin download a zip file of all report cards for a specific class with one click?
  - [ ] **Deep-Link Generation:** Instead of just a download, is there a "Copy Link" option to send a unique, secure PDF link directly to a parent via WhatsApp?
  - [ ] **Signature Overlay:** Ability for the Principal to upload a digital signature that appears automatically on every downloaded report card.
  - [ ] **Watermark Logic:** If the "Revenue Gate" is active, can the system prevent the download (or apply a "UNPAID" watermark) if the fee balance is not zero?
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